

1. What is the primary function of a router in a computer network?

- a) Assigning IP addresses to devices
- b) Providing wireless connectivity to devices
- c) Forwarding data packets between networks
- d) Managing user authentication and access control

Answer – c) Forwarding data packets between networks

2. What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?

- a) Assigning static IP addresses to devices
- b) Resolving domain names to IP addresses
- c) Managing network traffic and congestion
- d) Dynamically assigning IP addresses to devices

Answer - d) Dynamically assigning IP addresses to devices

3. Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?

- a) Router
- b) Switch
- c) Hub
- d) Repeater

Answer - b) Switch

4. Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?

- a) Star
- b) Bus
- c) Ring
- d) Mesh

Answer - b) Bus

True or False: A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain.

Answer - True

True or False: TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.

Answer – False

True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

Answer - True

8. Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.

Answer -

10. Discuss the importance of network documentation in the context of building and managing networks.

1. Ans - Client Configuration

First, we check the setup or requirements of the client. This helps us understand what kind of network they need – big or small, simple or complex.

2. Choosing the Right Topology

Then we decide the network topology – like star, bus, or mesh –

depending on how the computers will be connected. The right topology makes the network work better.

3. What Cables and Devices to Use

We also decide what type of cables (like fiber or LAN) and devices (routers, switches) we'll need. This depends on the speed, distance, and budget.

4. Setting IP Adresse

We plan the IP address system – who gets which address, how we divide them , and whether they'll be fixed or automatic . This avoids confusion and keeps the network organized.

5. Router and Switch Setup

We document how routers and switches are configured for example, what settings are used, which ports, and how security is set up.

6. Area Size

We also think about how big the place is. it's a small office or a large building, we plan the cable length, where to place devices, and how to get full Wi-Fi coverage.

7. Cost-Effective Planning

Good And Cheap helps save money.